

QUESTION1) The goal is to make a model of the DC motor and see how it moves when a short voltage is given.

SOLUTION1) In this case, The system shows an **overdamped response** with a smooth decay in speed after the impulse. when there is no load the mechanical part of the motor is dominant. Since the electrical effects are instantaneous (impulse), the motor's behavior then becomes mechanical free decay. Then our equation:

$$\frac{b}{s^2 + a_1 s + a_0} = \frac{b}{(s + p_1)(s + p_2)} \quad (1.1) \quad \frac{100b}{0.75} = 20 \quad b = 0.15 \quad (1.3)$$

$$G(s) = \frac{0.15}{s^2 + 3.25s + 0.75} \quad (1.2) \quad 6\tau = 24 \quad \tau = 4 \quad \tau = 1/p_1 \quad (1.4)$$

$$p_1 = 0.25 \quad p_2 = 12p_1 \quad p_2 = 3$$

Justify;

The pole closest to the origin determines the system's response rate and is called the dominant pole. In this case, p_2 is 12 times farther than p_1 . The system's 24-second rest time is considered the setting time, which is consistent with the 6τ equation. This demonstrates that the pole placement is logical and that the system's behavior is determined by the p_1 pole.

```
s = tf('s');
a2 = 1;
a1 = 3.25;
a0 = 0.75;
b = 0.15;
d = [a2 a1 a0];
tf2 = tf(n,d);
tf2 = (0.15) / (s^2 + 3.25*s + 0.75);
n = roots(d);
[p,t] = impulse(tf2);
scale = 20 / max(y);
plot(t, y*scale, 'b', 'linewidth', 0.1);
hold on;
yline(20, 'r--', 'Target max speed');
xlabel('Time (s)');
ylabel('Angular speed w(t) [rad/s]');
title('DC Motor with mechanical part');
legend('DC motor speed response', 'Target max speed');
grid on;
```

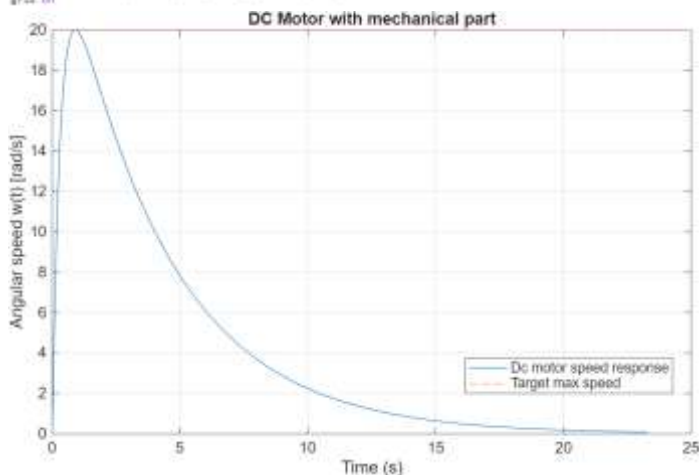


Figure 1.1 DC MOTOR WITH MECHANICAL PART

In this graph, when a sudden voltage is applied to a DC motor the motor rapidly accelerates to 20 rad/s and then stops after 24 seconds due to damping. The system is overdamped and stable showing a realistic mechanical decay, indicating excessive damping because the speed response gradually decreases. The model is consistent with physical behavior.

QUESTION2) The goal is to make a model of the conveyor and see how it moves with a constant torque. The conveyor is a second-order system. It moves slowly after a short delay and then stop.

SOLUTION2) In this situation, 10 torque applied means the initial value is zero, an input that remains constant. It's a step signal. Despite the input, the input doesn't respond for 2 seconds and remains zero. The 6-tau rule should be applied to ensure the system reaches its final value within 6 seconds after a 2-second delay. Then equation:

τ -time constant, K -gain, $G_c(s)$ -TF of the conveyor, τ_d -delay
 $T(s)$ -tork, $h(t)$ -unit step function

$$\tau_d = 2s \quad \text{6}\tau \text{ rule,} \quad T(t) = 10 \cdot h(t) \quad (2.2)$$

$$K = \frac{10N}{1rad/s} = 10 \quad 6\tau = 6s \quad T(s) = \frac{10}{s}$$

$$\tau = 1s$$

$$G_c(s) = \frac{K e^{-\tau_d s}}{s+1} = \frac{(10)e^{-2s}}{s+1} \quad 1 = \lim_{s \rightarrow 0} s \cdot \left(\frac{K}{s+a} \right) \cdot \left(\frac{10}{s} \right) \quad (2.3)$$

```
s = tf('s');
K = 10;
tau_d = 2;
tau_d = 2;
n1 = K*10;
d1 = [1 1];
tf1 = exp(-tau_d*s)*tf(n1,d1);
step(tf1);
xlabel('Time (s)');
ylabel('Angular speed (rad/s)');
title('Conveyor response to 10 N torque (First order + delay)');
grid on;
```

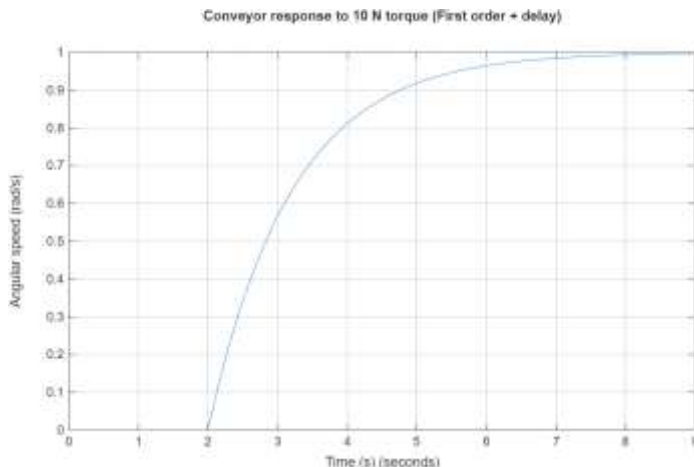


Figure 2.1 CONVEYOR RESPONSE TO 10N TORQUE (FIRST ORDER + DELAY)

In this graph, the conveyor remains stationary for the first 2 seconds, then gradually increases its angular speed and reaches a steady-state value around 1 rad/s after the delay.

QUESTION3) The goal is to show the gear problem that changes the speed by 2 rad/s every 2 seconds.

SOLUTION3) In this case, The system output shows periodic fluctuations in speed due to the gear defect. The term "rotational speed reduction" indicates that this defect affects the system's final output speed that is, its angular velocity (rad/s). The fact that it decreases every two seconds indicates periodicity. It is modelled with an impulse function. This is called an impulse train. There is a decrease in the signal. Then equation:

t -time, d_0 -output disturbance, k -counter

$$d_o(t) = -2 \sum_{k=1}^{\infty} \delta(t-2k) \quad (3.1) \quad d_o(s) = -2 \cdot \sum_{k=1}^{\infty} \mathcal{L}\{\delta(t-2k)\} \quad (3.3)$$

$$d_o(s) = -2 \left(\frac{e^{-2s}}{1 - e^{-2s}} \right) \quad G(s) = \left[\frac{0.15}{s^2 + 3.25s + 0.75} V(s) \right] - \sum_{k=1}^{\infty} 2e^{-2ks} \quad (3.4)$$

```
s = tf('s');
G = (0.15)/(s^2 + 3.25*s + 0.75);
[y, t] = impulse(G);
y = ctrapz(t, y);
y = 20*y/max(y);
y_dist = y - 2*floor(t/2);
figure;
plot(t, y, 'b--', 'linewidth', 1.5);
hold on;
plot(t, y_dist, 'r-', 'linewidth', 2);
title('Periodic Speed Rise Effect');
xlabel('Time (s)');
ylabel('Angular speed (rad/s)');
legend('Normally', 'Disturbed');
grid on;
```

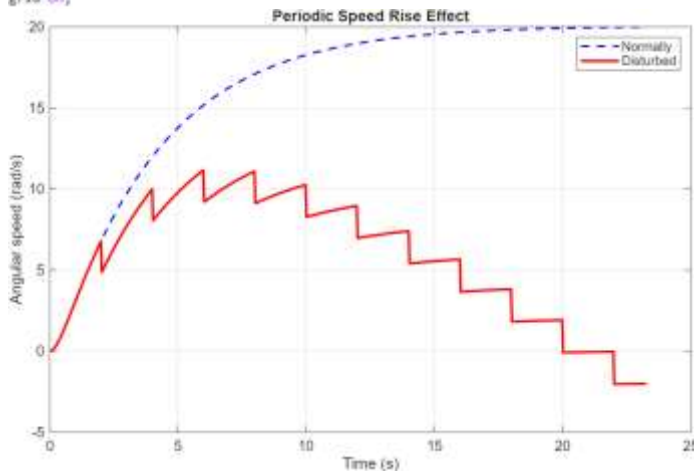


Figure 3.1 PERIODIC SPEED RISE EFFECT

QUESTION4) The goal is to show the roller that always reduces the torque by 1 N. The system has a small steady-state error because of the torque loss.

SOLUTION4) At this point, 1N is mentioned and the newton is the unit of torque. It affects the torque input to the conveyor system in question 2. This type falls into the internal parametric disturbance category. Constants indicate a step signal that starts at zero. Therefore, an undesirable torque is present, and an integral must be used to correct the structure resulting from these disturbances. It can combine small errors and tolerate friction. Then equation:

$d_i(t)$ -disturbance tork

$$d_i(t) = -1 \cdot h(t) \quad (h(t) \text{ is the unit step function}) \quad (4.1)$$

$$d_i(s) = \mathcal{L}\{d_i(t)\} = -\frac{1}{s} \quad (4.2)$$

$$T_{net}(s) = T_{motor}(s) + d_i(s) = T_{motor}(s) - \frac{1}{s} \quad (4.3)$$

$$Z_{out}(s) = G_c(s) \cdot T_{net}(s) \quad (4.4)$$

$$Z_{out}(s) = \left[\frac{(0.1)e^{-2s}}{s+1} \right] \cdot \left(T_{motor}(s) - \frac{1}{s} \right) \quad (4.4)$$

$$\text{motor torque: } T_{motor}(t) = 10N \Rightarrow T_{motor}(s) = \frac{10}{s} \quad (4.5)$$

$$\text{disturbance torque: } d_i = -1N \Rightarrow d_i(s) = -\frac{1}{s} \quad (4.6)$$

$$\omega_{ss} = \lim_{s \rightarrow 0} s \cdot \left[\left[\frac{(10)e^{-2s}}{s+1} \right] \cdot \left(\frac{10}{s} - \frac{1}{s} \right) \right] \quad \omega_{ss} = 0.1 \cdot 9 = 0.9 \text{ rad/s} \quad (4.7)$$

$$\omega_{ss} = \lim_{s \rightarrow 0} s \cdot \left[\left[\frac{(10)e^{-2s}}{s+1} \right] \cdot \left(\frac{9}{s} \right) \right] \quad \omega_{ss} = \lim_{s \rightarrow 0} s \cdot \left[\frac{0.1 \cdot (1)}{(1)} \right] \cdot \left(\frac{9}{s} \right) \quad (4.8)$$

A permanent loss of 1N is a torque loss, which directly affects the steady-state state of the system. According to the viscous equation $T=kW$, $10N=k \cdot 20\text{rad/s}$, $k=0.5\text{Ns/rad}$. T_{net} decreases to $9N$ $9N=0.5w_{new}$, $w_{new}=18\text{rad/s}$. This analysis proves that the steady-state error is permanent.

```
s = tf('s');
K = 10;
tau = 1;
tau_d = 2;
d1 = [1 1];
n1_i = K * 10;
n1_d1 = K * 9;
tf_i = tf(n1_i, d1, 'delay', tau_d);
tf_d1 = tf(n1_d1, d1, 'delay', tau_d);
figure;
step(tf_i, 'b--', tf_d1, 'r-');
xlabel('Time (s)');
ylabel('Angular speed(rad/s)');
title('Conveyor response ideal vs disturbed');
legend('Ideal response SS 1.0 rad/s', 'Disturbed response SS 0.9 rad/s');
grid on;
```

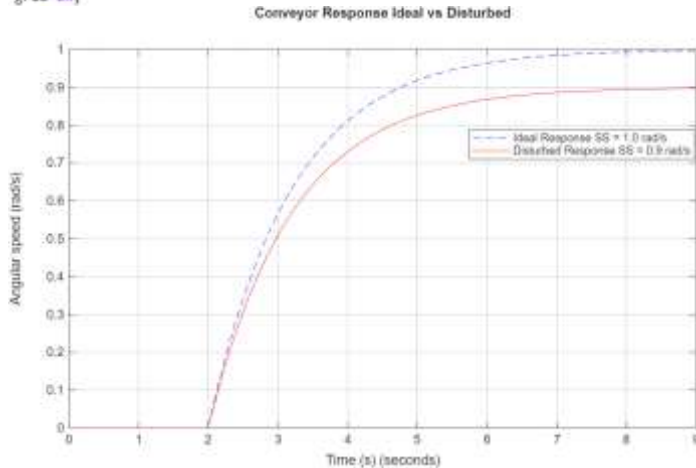
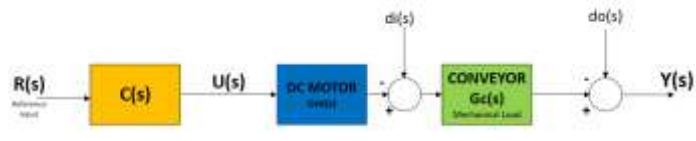


Figure 4.1 CONVEYOR RESPONSE IDEAL VS DISTURBED

In this graph, the permanent difference of 0.1 rad/s between 1.0 rad/s and 0.9 rad/s is the steady state error due to the 1N torque loss.

QUESTIONS The goal is to draw the full block diagram of the motor and conveyor system.

SOLUTIONS The block diagram, the model represents an open-loop control system including both input and output uncertainties, begins with the reference input. The controller converts the reference into a control signal. The control signal feeds the system's actuator, block $G_m(s)$. The DC motor converts this voltage into torque, driving the $G_c(s)$ load. This mechanical load ultimately produces the final speed. The diagram includes two main uncertainties faced by the system: one is the internal disturbance, $d_i(s)$, and the other is the external disturbance $d_o(s)$. $d_i(s)$ is positioned as the input disturbance, where the torque loss is subtracted from the motor's generated torque. $d_o(s)$ is defined as the output disturbance, where the 2 rad/s drop is subtracted from the conveyor's final speed. $d_i(s)$ is **parametric** because it changes the system parameters such as input torque or gain $d_o(s)$ is **non-parametric** because it directly affects the output signal without changing the system parameters.



$R(s)$ - Reference Input $U(s)$ - Controller output $d_i(s)$ - internal disturbance
 $d_o(s)$ - external disturbance $Y(s)$ - System output $T_m(s)$ - Motor torque
 $C(s)$ - Controller TF $G(s)$ - Plant TF $G_m(s)$ - DC motor TF
 $G_c(s)$ - Conveyor TF $T_{net}(s)$ - Net torque $W_c(s)$ - Conveyor speed

$$U(s) = C(s)R(s) \quad T_m(s) = G_m(s) \cdot U(s) = G_m(s) \cdot C(s) \cdot R(s) \quad (5.1)$$

$$T_{net}(s) = T_m(s) - d_i(s) = [G_m(s) \cdot C(s) \cdot R(s)] - d_i(s)$$

$$W_c(s) = G_c(s) \cdot T_{net}(s) = G_c(s) \cdot [G_m(s) \cdot C(s) \cdot R(s)] - d_i(s)$$

$$W_c(s) = [G_c(s) \cdot G_m(s) \cdot C(s) \cdot R(s)] - [G_c(s) \cdot d_i(s)]$$

$$Y(s) = W_c(s) - d_o(s)$$

$$Y(s) = \underbrace{[C(s)G_m(s)G_c(s)]R(s)}_{\text{Reference input effect}} - \underbrace{[G_c(s)d_i(s)]}_{\text{The effect of internal disturbance}} - \underbrace{d_o(s)}_{\text{The effect of output disturbance}}$$

QUESTION6 The goal is to find the open-loop equation and use a simple gain controller for 15 rad/s speed.

SOLUTION6 The open-loop system with a pure gain controller reaches the target speed but it is not very stable because it has no feedback. The controller motor and the conveyor are connected in series. Therefore TF must be the product of these blocks.

$$G_{OL}(s) = C(s) \cdot G_m(s) \cdot G_c(s) \quad (6.1)$$

$$G(s) = \frac{0.15}{s^2 + 3.25s + 0.75} \quad (6.2)$$

$$G_{OL}(s) = K_p \cdot \left[\frac{0.15}{s^2 + 3.25s + 0.75} \right] \cdot \left[\frac{(10) \cdot e^{-2s}}{s+1} \right] \quad (6.3)$$

$$Y_{ss} = \lim_{t \rightarrow \infty} y(t) = \lim_{s \rightarrow 0} s \cdot Y(s) \quad (6.4)$$

Knowing that the system is stable, the Final Value Theorem is used to find the steady state output at infinite time t.

$$15 = \lim_{s \rightarrow 0} s \cdot \left[G_{OL}(s) \cdot \left(\frac{1}{s} \right) \right] \quad (6.5)$$

$$15 = \lim_{s \rightarrow 0} G_{OL}(s)$$

$$15 = \lim_{s \rightarrow 0} \left(K_p \cdot G_m(s) \cdot G_c(s) \right)$$

$$15 = K_p \cdot \left(\lim_{s \rightarrow 0} G_m(s) \right) \cdot \left(\lim_{s \rightarrow 0} G_c(s) \right)$$

By substituting the gains into the equations, the pure gain that will provide 15 rad/s is found.

$$G_m(0) = \lim_{s \rightarrow 0} \left[\frac{0.15}{s^2 + 3.25s + 0.75} \right] = \frac{0.15}{0+0+0.75} = 0.2 \quad (6.6)$$

$$G_c(0) = \lim_{s \rightarrow 0} \left[\frac{(10)e^{-2s}}{s+1} \right] = \frac{(10) \cdot 1}{0+1} = 10$$

$$K_p = \frac{15}{20} = \frac{3}{4} = 0.75$$

$$G_{OL}(s) = \frac{(0.75) \cdot (0.15) \cdot (10) \cdot e^{-2s}}{(s^2 + 3.25s + 0.75)(s+1)} = \frac{(1.125) \cdot e^{-2s}}{(s^2 + 3.25s + 0.75)(s+1)} \quad (6.7)$$

When a unit step reference input of 1 unit (1 Volt) is applied to an open loop system, the steady state output of the system will be 15 rad/s.

```
s = tf('s');
Kp = 0.75;
Gm = (0.15) / (s^2 + 3.25*s + 0.75);
Gc = (10) * exp(-2*s)/(s+1);
n = 10;
d = [1 1];
Gc = tf(n, d, 'InputDelay', 2);
G_OL = Kp * Gm * Gc;
figure;
step(G_OL);
title('Open Loop Step Response Kp = 0.75');
ylabel('Angular speed (rad/s)');
xlabel('Time (s)');
yline(15, 'r--', 'target speed = 15 rad/s');
grid on;
```

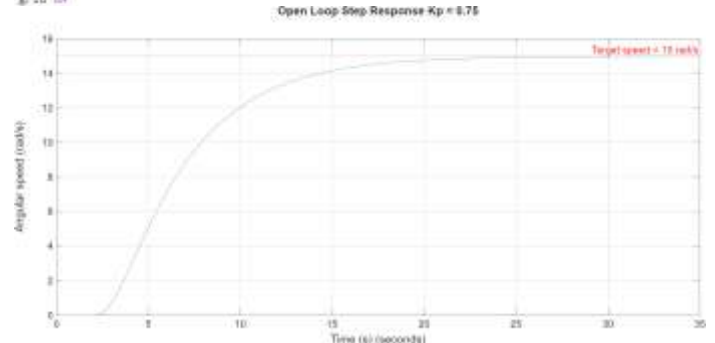


Figure 6.1 OPEN LOOP STEP RESPONSE

In this graph, due to the 2-second delay there is a 2-second wait before the motor starts. After the 2-second delay, the motor slowly increases due to the effect of the slow dominant pole, reaching a steady state at 15 rad/s thanks to the gain.